

Minireview

Energetics of kinesin-1 stepping mechanism

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Abstract Kinesin-1 is a dimeric motor protein that transports cellular cargo along microtubules by using the energy released from ATP hydrolysis and moving processively in 8-nm steps. Recent novel studies at the single molecular level have provided extensive knowledge on how kinesin-1 converts the free energy of ATP hydrolysis and uses it for “walking” along microtubules. In this review, I have discussed the important topics pertaining to the energetics of kinesin-1 stepping mechanism and the consensus walking model.

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1. Introduction

Kinesin-1, formerly called conventional kinesin, belongs to a family of motor proteins that are associated with the microtubule cytoskeleton. The kinesin-1 molecule consists of two heavy chains (usually called heads) that are responsible for motor activity, and two light chains that are involved in cargo association [1]. It is a processive motor and a single-molecule walker that can take hundreds of sequential steps along a microtubule under load [1–4]. With each step, the two kinesin heads exchange leading and trailing positions as they alternately hydrolyze ATP, generating a “hand-over-hand” motion [5,6]. Thus far, three main models have been proposed to describe the mechanism underlying kinesin-1 movement, namely, inchworm, symmetric hand-over-hand, and asymmetric hand-over-hand models. Recent biophysical experiments favor the asymmetric hand-over-hand model [4,5,7,8]. However, the mechanism responsible for the remarkable processivity of kinesin remains unclear. Furthermore, the details of how the chemical kinetic cycle couples to the mechanical cycle are unclear. In this review, I describe the topics related to the energetics of the stepping mechanism of kinesin-1, the current consensus walking model, and the future prospects in this field of study.

2. Energetics of the unidirectional and biased motion of kinesin-1

The neck-linker is a short flexible sequence (consisting of approximately 15-amino-acids) that links the kinesin head to

its neck, and reversibly docks the main part of the head. The neck-linker docking model proposed by Rice et al. [9] is seminal. The neck-linker is an essential component of kinesin-1, in the absence of which the stepping behavior of kinesin-1 is not observed [10]. A number of quantitative models for the stepping behavior of kinesin-1 have been developed taking the proposals of the neck-linker model as a starting point. It was originally proposed that neck-linker docking is the force-generating engine of kinesin-1 [11], however, the displacement of the neck-linker region (~ 2 nm) alone cannot explain the observed size of the kinesin-1 step (~ 8 nm) as discussed in recent reviews [4,5,12,13]. Furthermore, the free energy associated with neck-linker docking is only approximately 5% of that released through the hydrolysis of one ATP molecule (50–60 kJ/mol, or $\sim 22 k_B T$) [11]. Taking into account the fact that the efficiency of kinesin-1 is approximately 50% [4], a mismatch is observed between the energy released by neck-linker docking and that released through ATP hydrolysis. Interestingly, Hackney [14] demonstrated using an oxygen isotope that the free energy released during the binding of the ATP on the kinesin head to a microtubule was 9–13 $k_B T$, which is equivalent to that required to drive a complete 8-nm step against a stall force of 7 pN. Furthermore, Bier [15] theoretically estimated that the amount of energy required to drive the power stroke was 11–13 $k_B T$. These values support the finding that efficiency of kinesin-1 was 50%.

From these results, which are in fair agreement with those related to the observed energetics, it is apparent that some mechanism other than neck-linker docking is needed to generate the force for kinesin-1 stepping at high load. Because of the small size of the head and the large step size of kinesin-1, it is believed that diffusion (i.e., Brownian ratchet) plays an important role in the transport of the head to the next binding site on the microtubule [4]. Taniguchi et al. [16] proposed that approximately 6 $k_B T$ of bias was favorable for a forward step (plus end of the microtubule). They inferred this by measuring the temperature dependence of the forward and backward stepping rates. This result was based on the evidence that a Brownian ratchet-based mechanism is the underlying basis of kinesin-1 walking. The other possibility that creates this bias towards the plus end of the microtubule is the structural aspect of the microtubule.

As discussed by Kikkawa [13], the results of cryo-electron microscopy analysis revealed that a tilted conformation is induced by binding to the forward site of the neck-linker, which introduces a forward bias. This theory is supported by the results of the measurement of the unbinding force, which shows that a larger force ($\sim 140\%$) is required to detach the kinesin-1

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head when it is pulled in the minus direction (the direction opposite to the motion of kinesin-1) than when it is pulled in the plus direction [17–20]. Furthermore, in the presence of ATP, kinesin binds cooperatively to microtubules [21], increasing the possibility of asymmetric structural changes in the microtubules. Therefore, it is reasonable to conclude that the occurrence of unidirectional biased movements involves the three following components: power stroke, Brownian ratchet, and structural components. The next challenge is to determine which of the successive kinesin steps is associated with energy-dependent conformational motions (i.e., power strokes) and with diffusion (i.e., Brownian ratchet).

3. New information about the ATP-waiting state: one head or two heads?

The conjecture of whether kinesin-1 is in a two-headed bound state or a one-headed bound state (the so-called “waiting state”) for a longer duration during the process of stepping processively is controversial [4,14,22–24]. As kinesin-1 is expected to assume the “waiting state” at low ATP concentrations, it is necessary to understand this state for the elucidation of the walking mechanism. The interpretation of the data obtained from electron microscopy has long been debated. The results of image analysis by electron micrographic reconstruction of the dimeric kinesin-microtubule complex were inconclusive; either single-headed [25,26] or double-headed [27,28] binding was found to predominate both in the absence of added nucleotides (i.e., at low ATP concentrations) and in the presence of adenylyl imidodiphosphate (AMP-PNP), an ATP analogue. In solution, the binding stoichiometry (one kinesin dimer per tubulin heterodimer) yields a molar ratio of 2:1 in both the nucleotide-free state [29–31] and in the presence of AMP-PNP [29], implying single-headed binding. Block [4] argued that although these data are straightforward, these experiments were not carried out under physiological conditions, especially with respect to kinesin concentration. Hackney suggested that because the rear head of kinesin is competent in ATP synthesis, it necessarily remains bound to the microtubule for most of the kinesin cycle [14]. Furthermore, Yildiz et al. [32] directly observed the “waiting state” by fluorescently labeling one of the heads; this indicates a ~16-nm periodicity in the dwells, implying that the two heads are ~8-nm apart for most of the time. The observations also strongly support a model with both heads bound. On the other hand, the recent model proposed by Carter and Cross [12,23] suggests that kinesin is bound by a single head, whereas its partner head remains detached and free for diffusion between potential binding sites; thus, one-head bound is predominantly in the waiting state. Furthermore, they recently obtained experimental evidence that shows that in the “waiting state” the detached head has ADP [33]. My group measured the unbinding force and elastic modulus of a single kinesin-microtubule complex by using optical tweezers. The results suggest that two-headed binding is predominant in the presence of AMP-PNP, whereas single-headed binding is predominant in the absence of a nucleotide [17,20]. However, since our results were obtained under static condition, like the electron microscopic one, the possibility that one head may be weakly bound, whereas the other may be strongly bound, as pointed by Block [4], cannot be completely denied. Recently, the single-molecule

fluorescence resonance energy transfer (smFRET) analysis carried out by Mori et al. [34] resolved these contradictions; they suggested that either of the abovementioned theories may be correct depending on the conditions. They reported that when kinesin movements occur in the presence of saturated ATP concentration (physiological concentrations), both heads of kinesin remain bound to the microtubule for most of the time. However, when the nucleotide-binding process becomes the rate-limiting step at low ATP concentrations, kinesin waits for the ATP in a one-head bound state and makes brief transitions to a two-head bound intermediate state as it walks along the microtubule. This explains one of the contradictions described above. Nevertheless, several problems remain. As the current experimental instrumentation lacks temporal resolution, a more precise analysis of the transitions at high ATP concentrations is difficult. Hence, further technical breakthroughs are desired.

4. Consensus walking model of kinesin-1 and questions to be answered

Although considerable and impressive progress of the kinesin-1 stepping mechanism has been achieved, fundamental questions still abound. Therefore, it might be premature to attempt a consensus walking model of kinesin-1 stepping at this stage. Nevertheless, I describe below my proposal and discuss questions to be answered (Fig. 1).

First, in the absence of microtubules, ADP binds to both heads of kinesin-1. One head attaches to a microtubule and releases ADP on first encountering a microtubule, but the other head retains ADP [35]. The head that retains ADP either stays detached from the microtubule or attaches only weakly to the microtubule until the first head binds to ATP. The structural aspect of this state is controversial. On the basis of data from FRET analysis, Mori et al. [34] suggested that the detached head is positioned behind the nucleotide-free microtubule-bound head (left side of (A) in Fig. 1). On the other hand, Alonso et al. [33] proposed the gating model, in which they suggested that the detached head in this state is positioned in front of the bound head (right side of (A) in Fig. 1). Since this position is directly related to the “walking” mechanism, it is important that it is definitely identified. Carter and Cross [23] reported that the forward to backward step ratio varies exponentially with the load force, which suggests that increase in the backward load can progressively abrogate the forward bias in the stepping mechanism. From the perspective of energetics, a key advantage of the parking model of Alonso et al. [33] is that it places the “unbound” head close to its next binding site before ATP binding occurs, which accounts for the abovementioned suggestion of Carter and Cross [23]. According to the model proposed by Mori et al. [34], ATP binding is a necessary step for the shifting of the head forward from its trailing position by the docking of the neck-linker. The energetics of neck-linker docking suggests that docking cannot occur under load; however, efficient kinesin stepping is observed against substantial loads, and indeed, it is only efficient under loads. Further investigation in this regard is desired.

The binding of ATP to the microtubule-bound head stimulates neck-linker docking, including some conformational change. This change results in a mainly plus-end directed motion of the rear partner head through a comparatively small

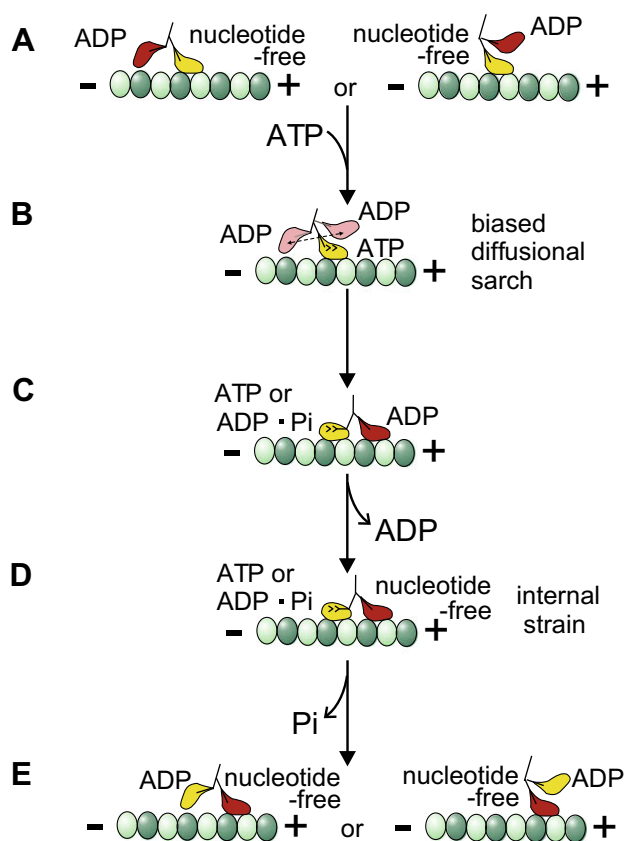


Fig. 1. In the absence of microtubules, ADP binds to both heads of kinesin-1. (A) After one head comes into contact with a microtubule, the head attaches to the microtubule and releases ADP, while the other head retains ADP and remains “detached” from the microtubule (transient interaction with the microtubule may occur). The position of the “detached” head remains undetermined (see text). (B) ATP binding to the attached head results in neck-linker docking (indicated by >>), and a structural change enables a biased diffusional search by the detached head for the next forward binding site. (C and D) After the partner head has reached its forward binding site, ADP is released, and the previously weakly bound head strongly binds to the microtubule, thereby creating internal strain. This internal strain prevents the binding of ATP to the forward (red) head. (E) After hydrolysis of ATP and the release of phosphate by the rear (yellow) head, the internal strain is released, which allows the nucleotide-free front (red) head to rebind with ATP for the next step. The situation reverts to (A), except that the heads have exchanged their relative positions, and the centroid of the molecule has advanced 8-nm along the microtubule.

displacement (approximately 1–2 nm). Following this state, the unbound partner head that has ADP, starts a biased diffusional search for its next forward binding site on the microtubule. At this time, the unbound partner head may go back to the previous backward binding site under a higher load [12]. The heads have now exchanged their relative positions, and the centroid of the molecule has advanced by 8-nm along the microtubule, which is the tubulin dimer repeat distance. After the partner head has reached its forward binding site, ADP is released, and the state of this partner head is transformed from a weak to a strong microtubule-binding state leading to internal strain [14,19]. The assumption that nucleotide dissociation occurs only when a head is in the forward position is supported by the findings of Guydosh and Block [36]. The results from the FRET data also support this assumption [34].

The strain generated by this double-headed binding state (both heads bind tightly to the microtubule) tends to prevent the early binding of ATP to the front head until after the rear head has had a chance to hydrolyze its own ATP and release phosphate. The strain is released after the release of phosphate from the rear head. This allows the nucleotide-free front head to rebind with the ATP for the next step. In other words, the internal strain that is generated should regulate the detachment of ADP from the tethered head [19]. It is reasonable to say that the position of the neck-linker also regulates the detachment of ADP from the tethered head (inhibits ADP release in the “docked” position), as suggested by Mori et al. [34].

As described above, some details of the stepping mechanism remain unknown and further studies are needed to complete the picture.

5. Future prospects with regard to the energetics of stepping

We have not yet completely understood the mechanism underlying kinesin-1 stepping and its energetics, especially, the possible transitions and intermediate positions during rapid stepping. For example, it remains uncertain whether kinesin-1 movements involve substeps within the cycle [23,37]. No theory for substeps has been established as yet [4,12,23]. However, it would not be contradictory to consider that some stopping points (i.e., substeps) exist as the stepping motion of kinesin-1 is based on Brownian motion. Because the existence of substeps is directly linked to its mechanism and energetics, it is essential to be elucidated. A particular daunting challenge is the FRET analysis of kinesin-1 under load. The elucidation of the details of backward stepping movements in response to loads will enable more precise determination of the energetics of kinesin-1 movement. Another challenge is to understand how the kinesin-1 molecules interact in a group of molecules. Recently, experiments on the movements of beads that were pulled by several kinesin-1 molecules demonstrated that the effect of random walking (i.e., Brownian motion) weakens with an increase in the number of interactions with the microtubules [38]. Knowledge of the actual number of kinesin-1 molecules that interact with cellular cargo and microtubules will facilitate a more in-depth analysis of the movement of kinesin-1 *in vivo* and its energetics.

The interaction between kinesin and microtubules has been implicated in several serious diseases, which is one reason why medical scientists are particularly interested in revealing the finer details of how kinesin-1 moves across microtubules. Complete elucidation of the mechanism of the movement of kinesin-1 may open new avenues for investigations in the treatment of neurodegenerative diseases, such as hereditary spastic paraplegia that are caused by mutations in the neuronal kinesin-1 gene (KIF5A) [39]. Furthermore, recent results that post translational modifications of the microtubules influence the recruitment of kinesin-1 to a specific microtubule track *in vivo* suggest that some microtubule modifications are important markers of distinct microtubule populations and control kinesin-1 trafficking [40]. In this case, the modifications should alter the energy potential of the microtubules. Energy compliance is essential to account for the overall biological phenomenon.

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